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## Circuits and method for controlling a synchronous rectifier


### Abstract

A method and circuit controls a synchronous rectifier. The synchronous rectifier is turned OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero. One or more values of the drain or the source of the synchronous rectifier are monitored. The gate of the synchronous rectifier is driven to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below one or more threshold values.



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## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 6107844 | 12/1999 | Berg et al. | N/A | N/A |
| 10886853 | 12/2020 | Xu | N/A | H02M 1/08 |
| 2001/0019278 | 12/2000 | Sudo et al. | N/A | N/A |
| 2008/0084197 | 12/2007 | Williams et al. | N/A | N/A |
| 2008/0094124 | 12/2007 | Happenstall et al. | N/A | N/A |
| 2010/0271117 | 12/2009 | Chou | N/A | N/A |
| 2012/0087167 | 12/2011 | Kuzumaki et al. | N/A | N/A |
| 2015/0171855 | 12/2014 | Braun | N/A | N/A |
| 2017/0207782 | 12/2016 | Takazawa | N/A | N/A |
| 2019/0140630 | 12/2018 | Chen | N/A | H03K 17/163 |
| 2019/0260368 | 12/2018 | Vytla et al. | N/A | N/A |
| 2021/0058001 | 12/2020 | Dou | N/A | H02M 3/1588 |
| 2021/0091668 | 12/2020 | Buhari et al. | N/A | N/A |
| 2022/0352812 | 12/2021 | Teggatz | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 111404526 | 12/2019 | CN | N/A |
| 2005328668 | 12/2004 | JP | N/A |
| 2006013540 | 12/2005 | WO | N/A |

**OTHER PUBLICATIONS**

R. Elferich and T. Lopez, “Impact of gate voltage bias on reverse recovery losses of power MOSFETs,” Twenty-First Annual IEEE Applied Power Electronics Conference and Exposition, 2006. APEC '06., Dallas, TX, USA, 2006, pp. 13890-1395 (Year: 2006). cited by examiner
Hoang, Ngan K., Sang-Gug Lee and Young-Jin Woo. “A zero current detector with low negative inductor current using forced freewheel switch operation in synchronous DC-DC converter,” 2014 IEEE Fifth International Conference on Communications and Electronics (ICCE), 2014, pp. 318-322. (Year: 2014). cited by examiner
International Search Report and Written Opinion from the USPTO for PCT/US2022/27377 dated Aug. 16, 2022, 18 pp. cited by applicant
Elferich, R., et al., “Impact of gate voltage bias on reverse recovery losses of power MOSFETs,” Twenty-First Annual IEEE Applied Power Electronics Conference and Exposition, 2006, APEC

2006, Dallas, TX, USA, pp. 1390-1395. cited by applicant
European Search Report of European Patent Application No. 22799393.8 dated Feb. 28, 2025, 10 pp. cited by applicant

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part application of U.S. patent application Ser. No. 17/735,208 filed on May 3, 2022, which claims priority to U.S. Provisional Application No. 63/183,286 filed on May 3, 2021, the contents of each of which are incorporated by reference herein.

STATEMENT OF FEDERALLY FUNDED RESEARCH
(1) Not applicable.
TECHNICAL FIELD OF THE INVENTION
(2) The present invention relates in general to power switching applications. In particular, the present invention relates to reducing or eliminating current and voltage transients and the negative effects of current and voltage transients in power switching applications.
BACKGROUND OF THE INVENTION
(3) Reverse recovery charge is a challenge in all power switching applications, for example, motor control, solenoid control, or power management. Reverse recovery charge is stored at the junction of the body diode of a switch in a power switching application when the body diode is being forward biased. This charge increases as the forward bias current increases. Ideally, when a forward biased diode is suddenly put into reverse bias there will be no current flow in the reverse bias. However, with reverse recovery charged stored, when the forward biased diode is suddenly put into the reverse bias there will be a reverse current flow which will be a function of the charge that was stored when operating in the forward bias. The current and voltage transients caused by this reverse recovery charge cause a number of problems in power switching applications.
SUMMARY OF THE INVENTION
(4) The following embodiments reduce or eliminate current or voltage transients in the switching circuit by actively monitoring the SW pin located between the switching transistor and the synchronous rectifier. The circuits use closed loop feedback to control level transistion(s) for the synchronous rectifier gate. The active feedback may include a voltage, a rate of change of voltage, a current, a rate of change of current or a combination thereof, all at the SW pin. The circuit does not require, but may use, signals from or monitoring of the switching transistor. Moreover, the circuit does not require, but may include, a time delay circuit. As a result, the circuits and methods described herein reduce current losses from break before make, gate to drain capacitance, reverse recovery charge (Qrr) and gate bounce. The methods and circuits also help prevent ringing.
(5) In one embodiment of the present disclosure, a method for controlling a synchronous rectifier is provided. The synchronous rectifier is turned OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero. One or more values of a drain or a source of the synchronous rectifier are monitored. The gate of the synchronous rectifier is driven to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below one or more threshold values.

(6) In one aspect, a switching circuit includes the synchronous rectifier connected to a switching transistor, and the method further includes reducing or eliminating current or voltage transients in the switching circuit. In another aspect, the method further includes selecting the one or more first voltages or the second voltage. In another aspect, the one or more first voltages are less than a threshold voltage of the synchronous rectifier. In another aspect, the one or more first voltages include a set of descending stair-step values, or a set of descending values forming a ramp or a curve. In another aspect, the one or more values can be a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier. In another aspect, the method further includes adjusting the one or more threshold values or a delay in driving the gate of the synchronous rectifier to the second voltage using a predictive feedback process. In another aspect, the predictive feedback process samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier, and uses the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the one or more threshold values. In another aspect, the predictive feedback process samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and uses the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the one or more threshold values. In another aspect, the predictive feedback process samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier, and uses the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the predictive feedback process samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and uses the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the second voltage is equal to zero volts. In another aspect, the second voltage is less than zero volts. In another aspect, a gate of a switching transistor connected to the synchronous rectifier is not monitored or used to control the gate of the synchronous rectifier.

(7) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a driver circuit connected to the gate of the synchronous rectifier; a parallel clamping circuit connected to the gate of the synchronous rectifier; a monitoring circuit connected to the drain or the source of the synchronous rectifier and the parallel clamping circuit; and a control circuit connected to the driver circuit and the monitoring circuit. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the driver circuit includes: a first transistor having a gate connected to the control circuit and a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a gate connected to the control circuit; a one-way current switch connected to a source of the second transistor. In another aspect, the parallel clamping circuit includes a third transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the monitoring circuit and a source connected to a reference ground. In another aspect, the monitoring circuit includes: a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier, a gate connected to the control circuit, and a source connected to the parallel clamping circuit; and a fifth transistor having a drain connected to the parallel clamping circuit and a gate connected to the control circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the driver circuit, the parallel clamping circuit, the monitoring circuit and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

(8) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier;

a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connected between a source of the second transistor and a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the one or more first voltages by: turning the first transistor from ON to OFF; turning the second transistor from OFF to ON; turning the fourth transistor from OFF to ON; and turning the fifth transistor from ON to OFF. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the second voltage by turning the third transistor from OFF to ON. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the first transistor, the second transistor, the one-way current switch, the third transistor, the fourth transistor, the fifth transistor and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.
(9) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a body diode, a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.
(10) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.
(11) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connected to a source of the second transistor; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current

switch and a source connected to a reference ground; a second one-way current switch connected between a voltage and the gate of the fourth transistor; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a seventh transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

(12) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(13) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(14) In one embodiment of the present disclosure, a circuit includes a synchronous rectifier driver circuit. The synchronous rectifier driver circuit is configured to: turn a synchronous rectifier OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero, monitor one or more values of a drain or a source of the synchronous rectifier, and drive the gate of the synchronous rectifier to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below one or more threshold values.

(15) In one aspect, the synchronous rectifier driver circuit reduces or eliminates current or voltage transients in a switching circuit including the synchronous rectifier connected to a switching transistor. In another aspect, the one or more first voltages are less than a threshold voltage of the synchronous rectifier. In another aspect, the one or more first voltages include a set of descending stair-step values, or a set of descending values forming a ramp or a curve. In another aspect, the one or more values can be a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier. In another aspect, the one or more threshold values or a delay in driving the gate of the synchronous rectifier to the second voltage are adjusted using a predictive feedback process. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous

rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the one or more threshold values. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the one or more threshold values. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the second voltage is equal to zero volts. In another aspect, the second voltage is less than zero volts. In another aspect, a gate of a switching transistor connected to the synchronous rectifier is not monitored or used to control the gate of the synchronous rectifier.

(16) In another aspect, the synchronous rectifier driver circuit includes: a driver circuit connected to the gate of the synchronous rectifier; a parallel clamping circuit connected to the gate of the synchronous rectifier; a monitoring circuit connected to the drain or the source of the synchronous rectifier and the parallel clamping circuit; and a control circuit connected to the driver circuit and the monitoring circuit. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the driver circuit includes: a first transistor having a gate connected to the control circuit and a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a gate connected to the control circuit; a one-way current switch connected to a source of the second transistor. In another aspect, the parallel clamping circuit includes a third transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the monitoring circuit and a source connected to a reference ground. In another aspect, the monitoring circuit includes: a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier, a gate connected to the control circuit, and a source connected to the parallel clamping circuit; and a fifth transistor having a drain connected to the parallel clamping circuit and a gate connected to the control circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the driver circuit, the parallel clamping circuit, the monitoring circuit and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

(17) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connected between a source of the second transistor and a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the one or more first voltages by: turning the first transistor from ON to OFF; turning the second transistor from OFF to ON; turning the fourth transistor from OFF to ON; and turning the fifth transistor from ON to OFF. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the second

voltage by turning the third transistor from OFF to ON. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the first transistor, the second transistor, the one-way current switch, the third transistor, the fourth transistor, the fifth transistor and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

(18) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a body diode, a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.

(19) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier; a second one-way current switch connected from a source to the drain of the fourth transistor; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(20) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a body diode and a drain connected to the gate of the synchronous rectifier; a fourth transistor having a body diode, a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current switch and a source connected to a reference ground; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a seventh transistor having a drain connected to the drain or the source of the synchronous rectifier; a third one-way current switch connected from a source to the drain of the seventh transistor; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

(21) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth

transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.
(22) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.
(23) In one embodiment of the present disclosure, a power converter includes a switching transistor, a switching transistor driver circuit connected to a gate of the switching transistor, a synchronous rectifier connected the switching transistor, and a synchronous rectifier driver circuit connected to a gate of the synchronous rectifier and a drain or a source of the synchronous rectifier. The synchronous rectifier driver circuit is configured to: turn a synchronous rectifier OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero; monitor one or more values of a drain or a source of the synchronous rectifier; and drive the gate of the synchronous rectifier to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below one or more threshold values.
(24) In one aspect the synchronous rectifier driver circuit reduces or eliminates current or voltage transients in a switching circuit including the synchronous rectifier connected to a switching transistor. In another aspect, the one or more first voltages are less than a threshold voltage of the synchronous rectifier. In another aspect, one or more first voltages comprise a set of descending stair-step values, or a set of descending values forming a ramp or a curve. In another aspect, the one or more values can be a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier. In another aspect, the one or more threshold values or a delay in driving the gate of the synchronous rectifier to the second voltage are adjusted using a predictive feedback process. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the one or more threshold values. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the one or more threshold values. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the predictive feedback process is configured to sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a

switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the second voltage is equal to zero volts. In another aspect, the second voltage is less than zero volts. In another aspect, a gate of a switching transistor connected to the synchronous rectifier is not monitored or used to control the gate of the synchronous rectifier.
(25) In another aspect, the synchronous rectifier driver circuit includes: a driver circuit connected to the gate of the synchronous rectifier; a parallel clamping circuit connected to the gate of the synchronous rectifier; a monitoring circuit connected to the drain or the source of the synchronous rectifier and the parallel clamping circuit; and a control circuit connected to the driver circuit and the monitoring circuit. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the driver circuit includes: a first transistor having a gate connected to the control circuit and a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a gate connected to the control circuit; a one-way current switch connected to a source of the second transistor. In another aspect, the parallel clamping circuit includes a third transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the monitoring circuit and a source connected to a reference ground. In another aspect, the monitoring circuit includes: a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier, a gate connected to the control circuit, and a source connected to the parallel clamping circuit; and a fifth transistor having a drain connected to the parallel clamping circuit and a gate connected to the control circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the driver circuit, the parallel clamping circuit, the monitoring circuit and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.
(26) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a body diode and a drain connected to the gate of the synchronous rectifier; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the one or more first voltages by: turning the first transistor from ON to OFF; turning the second transistor from OFF to ON; turning the fourth transistor from OFF to ON; and turning the fifth transistor from ON to OFF. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the second voltage by turning the third transistor from OFF to ON. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the first transistor, the second transistor, the one-way current switch, the third transistor, the fourth transistor, the fifth transistor and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.
(27) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected

to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a body diode, a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.

(28) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a body diode and a drain connected to the gate of the synchronous rectifier; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(29) In another aspect, the synchronous rectifier driver circuit includes: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a body diode and a drain connected to the gate of the synchronous rectifier; a fourth transistor having a body diode, a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current switch and a source connected to a reference ground; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a seventh transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

(30) In another aspect, the synchronous rectifier driver circuit include: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a body diode and a drain connected to the gate of the synchronous rectifier; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(31) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the

second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the features and advantages of the present invention, reference is now made to the detailed description of the invention along with the accompanying figures, in which:

(2) FIG. **1**A shows a diode and the behavior over time of voltage and current in the diode. FIG. **1**B shows the behavior over time of current in the diode of FIG. **1**B at different transition times.

(3) FIG. **2**A shows a prior art power switching circuit. FIG. **2**B shows the behavior over time of behavior over time of voltage and current in the prior art power switching application shown in FIG. **2**A.

(4) FIG. **3**A shows the prior art power switching circuit of FIG. **2**A with parasitic inductances. FIG. **3**B shows the behavior over time of behavior over time of voltage and current in the prior art power switching application as shown in FIG. **3**A.

(5) FIG. **4** shows an embodiment of the present invention, a low switch of a power switching application.

(6) FIG. **5** shows the low switch of FIG. **4** in reverse-bias mode.

(7) FIG. **6** shows a resistance drive of the low switch of FIG. **5**.

(8) FIG. **7**A shows an embodiment of the present invention, a power switching circuit with a driver stage and a parallel clamping device with a time delay circuit. FIG. **7**B shows the behavior over time of voltages in the power switching circuit of FIG. **7**A.

(9) FIG. **8**A shows an embodiment of the present invention, a power switching application with driver stage, a parallel clamping circuit, and feedback monitoring. FIG. **8**B shows the behavior over time of voltages in the power switching circuit of FIG. **8**A.

(10) FIG. **9**A shows an embodiment of the present invention, a wide-bandgap low switch. FIG. **9**B shows a comparison of the behaviors over time of voltages in the switch of FIG. **9**A and in a prior art switch.

(11) FIG. **10** shows the equivalence of a four-terminal switch and a three-terminal switch with a body diode.

(12) FIG. **11**A shows an embodiment of the present invention, a four-terminal switch. FIG. **11**B shows the one type of behavior over time of voltages in the four-terminal switch of FIG. **11**A. FIG. **11**C shows another type of behavior over time of voltages in the four-terminal switch of FIG. **11**A.

(13) FIG. **12** shows an embodiment of the present invention, a power supply having a dynamic low side driver.

(14) FIG. **13** shows an embodiment of the present invention, the dynamic low side driver of FIG. **12**.

(15) FIG. **14** shows an embodiment of the present invention, signal timing diagrams for the power supply of FIG. **12**.

(16) FIG. **15** shows an embodiment of the present invention, flow chart of a method for controlling a synchronous rectifier.

(17) FIG. **16** shows an embodiment of the present invention, a block diagram of a power converter.

(18) FIG. **17** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.

(19) FIG. **18** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.

(20) FIG. **19** shows an embodiment of the present invention, a schematic diagram of a synchronous

rectifier driver circuit.
(21) FIG. **20** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.
(22) FIG. **21** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.
(23) FIG. **22** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.
(24) FIG. **23** shows an embodiment of the present invention, a schematic diagram of a synchronous rectifier driver circuit.
(25) FIG. **24** depicts signal diagrams of a buck switching regulator in accordance with the prior art.
(26) FIG. **25** depicts signal diagrams of a buck switching regulator in accordance with the prior art.
(27) FIG. **26** depicts signal diagrams of a buck switching regulator in accordance with the present invention using an active feedback circuit.
DETAILED DESCRIPTION OF THE INVENTION
(28) Illustrative embodiments of the system of the present application are described below. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
(29) In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the present application, the devices, members, apparatuses, etc. described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the device described herein may be oriented in any desired direction.
(30) Reverse recovery time is a challenge in all power switching applications, whether it is motor control, solenoid control, or power management. Reverse recovery charge is stored at the junction of the diode when it is being forward biased. (FIG. **1**A). The level of charge increases as the forward bias current increases. Ideally, when a forward biased diode is suddenly put into reverse bias there will be no current flow in the reverse bias. However, with reverse recovery charged stored, when the forward biased diode is suddenly put into the reverse bias there will be a reverse current flow, which will be a function of the charge that was stored when operating in the forward bias. If the transition time from forward bias to reverse bias is decreased the current peak from reverse recovery will increase. (FIG. **1**B).
(31) FIG. **1**A shows a diode **100** with a threshold voltage of V.sub.d and a forward current flowing through it (as denoted by the arrow above the diode **100**). Voltage **105** is the behavior of a voltage across the diode **100** over time as the voltage **105** is switched from a forward bias voltage V.sub.f to a reverse bias voltage V.sub.reverse. Ideal current I.sub.d **110** is the behavior of a current through the diode **100** as an ideal diode over time as the voltage **105** is switched from the forward bias voltage V.sub.f to the reverse bias voltage V.sub.reverse, with the ideal current I.sub.d **110** flowing while the voltage **105** is at V.sub.f and with the ideal current I.sub.d **100** not flowing while the voltage **105** is at V.sub.reverse. Practical current I.sub.d **115** is the behavior of a current through a realistic, practical diode **100** over time as the voltage **105** is switched from the forward bias voltage

V.sub.f to the reverse bias voltage V.sub.reverse, with the practical current I.sub.d **115** flowing while the voltage **105** is at V.sub.f and oscillating and diminishing to zero current when the diode **105** is switched to the reverse bias voltage V.sub.reverse because a reverse recovery charge Q.sub.rr is stored at the junction of the diode **100** while the practical current I.sub.d **115** is flowing while the voltage **105** is at the forward bias voltage V.sub.f and flows as a diminishing, oscillating transient current I.sub.d **115** for a short period after the switch from V.sub.f to V.sub.reverse. As shown in FIG. **1**B, the peak of the transient practical current I.sub.d increases as a transition time from V.sub.f to V.sub.reverse is decreased, with the practical current I.sub.d **120**, shown as a dashed line, having a relatively higher peak at a relatively short transition time and the practical current I.sub.d **125**, shown as a solid line, having a relatively lower peak at a relatively longer transition time.

(32) FIG. **2**A shows a prior art power switching circuit **200** such as that in a buck-switching regulator application, with a totem pole switch configuration including the high switch M**1 210** and the low switch M**2 225**. The high switch M**1 220** and the low switch M**2 225** are each shown as an exemplary metal-oxide semiconducting field-effect transistor (MOSFET) with a body diode, the parasitic diode that is intrinsic to the MOSFETs of the high switch M**1 220** and of the low switch M**2 225** as shown in FIG. **2**A. More particularly, the power switching circuit **200** includes the input for input power supply voltage V.sub.p **205**, the high switch M**1 210**, the input for the high gate voltage V.sub.gH **215**, the forward current I.sub.1 **220**, the low switch M**2 225**, the voltage source V.sub.dd **230** for the low gate voltage V.sub.gL **235**, the reverse current I.sub.2 **240**, the load inductor L.sub.1 **245**, the load capacitor C.sub.1 **250**, and the output for the output voltage V.sub.0 **255**.

(33) In the representative prior art power switching circuit **200**, to improve efficiency, the low switch M.sub.2 **225** is on the ON state when the current I.sub.2 is freewheeling through the load inductor L.sub.1 **245** and the load capacitor C.sub.1 **250**. When the output voltage V.sub.0 **255** decreases, more energy must be provided from the inpur power supply by inputting the input power supply voltage V.sub.p **205**, by switching the high switch M**1 210** to the ON state. To prevent a shoot-through current from the high switch M**1 210** to the low switch M.sub.2 **225**, the low switch M**2 225** must be switched to the OFF state before the high switch M**1 210** is switched to the ON state. The time period during which both the high switch M**1 210** and the low switch M**2 225** are in the OFF state is the deadtime. To switch the low switch M**2 225** to the OFF state, the voltage source V.sub.dd **230** is transitioned to 0 volts. During the deadtime that starts when the low switch M**2 225** is switched to the OFF state, a reverse recovery charge Q.sub.rr accumulates at the forward bias junction of the body diode of the low switch M**2 225**. When the high switch M**1 210** is switched to the ON state, ending the deadtime, the reverse recovery charge Q.sub.rr and the forward current I.sub.1 **220** flow through the high switch M**1 210**. To minimize the peak and the dI.sub.rr/dt of the reverse recovery charge Q.sub.rr that was accumulated in the body diode of the low switch M**2 225**, the transition time from the OFF state to the ON state can be increased as shown for the practical current I.sub.d **125** in FIG. **1**B, but at the cost of increased switching losses and a reduction in power efficiency.

(34) FIG. **2**B shows the behavior over time of the low gate voltage V.sub.gL **235**, shown as the low gate voltage V.sub.gL curve **260**; the high gate voltage V.sub.gH **215**, shown as the high gate voltage V.sub.gH curve **265**; and the current I.sub.1 **220** and the current I.sub.2 **240**, shown together as the current curve **275**. The current I.sub.1 flows as the low gate voltage is set to 0 V, beginning the deadtime. When the high gate voltage V.sub.gH **215** is set to switch the high switch M**1 210** to the ON state, the deadtime ends and the reverse recovery charge Q.sub.rr flows as the transient portion **270** of the current curve **275**.

(35) FIG. **3**A shows the representative prior art power switching circuit **200** with the high switch M**1 210**, the low switch M**2 225**, the current I.sub.1 **220**, the current I.sub.2 **240**, the parasitic inductance L.sub.p1 **280**, the parasitic inductance L.sub.p2 **285**, the parasitic inductance L.sub.p3 **290**, and the output switch pin voltage V.sub.sw **292** between the parasitic inductance L.sub.p2 **285**

and the parasitic inductance L.sub.p3 **290**. The product of (1) the sum L.sub.psum of the parasitic inductances L.sub.p1 **280**, L.sub.p2 **285**, and L.sub.p3 **290** and (2) the dI.sub.rr/dt of the reverse recovery charge Q.sub.rr that was accumulated in the body diode of the low switch M**2 225** translates to a parasitic inductance voltage transient. The high switch M**1 210** must be able to withstand the peak of this parasite inductance voltage transient, but as the breakdown voltage of the high switch M**1 210** is increased, the overall product of (1) the drain-to-source resistance R.sub.dsON of the high switch M**1 210** in the ON state and (2) the area of the high switch M**1 210** increases, and to keep efficiency lower, the size of the high switch M**1 210** must be increased. These considerations require trade-offs to be made in the types and sizes of the high switch M**1 210** and the low switch M**2 225**, management of board layout to minimize parasitics, and switching characteristics of the output of the representative prior art power switching circuit **200**.
(36) FIG. **3**B shows the behavior over time of the current I.sub.2 **240**, shown as the current I.sub.2 curve **295**, and the drain-to source voltage of the high switch M**1 210** V.sub.ds **297**, shown as the voltage V.sub.ds curve **299**. When the high switch M**1 210** is switched to the ON state, the deadtime ends, the reverse recovery charge Q.sub.rr flows as the transient portion **296** of the current I.sub.2 curve **295**, and the drain-to source voltage of the high switch M**1 210** V.sub.ds **297** increases, including the parasitic inductance voltage transient peak **298**.
(37) The solution to this problem is to eliminate or reduce the reverse recovery charge that can be stored. The elimination of reduction of the reverse recovery charge is achieved by gate controlling the turn-off voltage level of device M**2**. As shown in FIG. **4**, the gate of device M**2** is turned-off with a Vbias level. The Vbias is set to a voltage lower than the threshold voltage (Vt) of M**2**. Therefore, if a positive voltage were placed at the drain, no current, Ix, will flow through M**2** from drain to source.
(38) FIG. **4** shows an embodiment of the present invention, the low switch M**2 400** of a power switching application. The low switch M**2 400** includes a MOSFET with a body diode. The voltage source for the bias voltage V.sub.bias **405** is also shown and is connected to the input for the low switch gate voltage V.sub.gL **407**. The current I.sub.x is also shown. The low switch M**2 400** presents a solution to problems associated with the reverse recovery charge Q.sub.rr by gate controlling the turn-off voltage level of the low switch M**2 400**. The gate of the low switch M**2 400** is switched to the OFF state with a bias voltage V.sub.bias **405** by setting the bias voltage V.sub.bias **405** to a voltage lower than the forward threshold voltage V.sub.t of the low switch M**2 400**. If a positive voltage is placed at the drain of the low switch M**2 400**, the current I.sub.x **410** will be zero, i.e., no current I.sub.x **410** will flow from the drain to the source of the low switch M**2 400**.
(39) FIG. **5** shows the low switch M**2 400**, with the voltage source for the bias voltage V.sub.bias **405**, the gate-to-source voltage V.sub.gs **410** for the low switch M**2 400**, the reverse source voltage S.sub.rev **415**, and the reverse drain voltage D.sub.rev **420**, and the current I.sub.2 **425**. FIG. **5** also shows connected to the load inductor L.sub.1 **430**, the load capacitor C.sub.1 **435**, and the output for the output voltage V.sub.0 **440**. In the recirculation mode of a power switching device of which the low switch M**2 400** is a component, the current I.sub.2 will flow as shown, and in the low switch M**2 400**, the source and the drain reverse to become the reverse drain and the reverse source, respectively. The reverse drain voltage D.sub.rev **420** is shown grounded at 0 volts. The reverse threshold voltage of the low switch M**2 400** V.sub.t-reverse will be less than the forward threshold voltage V.sub.t because the backgate biasing, i.e., the bias voltage applied to the body diode, for the forward and reverse modes of the low switch M**2 400** are at different voltages with respect to the source voltage in the forward mode and the reverse source voltage S.sub.rev **415** for the reverse mode. With the reverse drain voltage D.sub.rev **420** grounded at 0 volts, the reverse source voltage S.sub.rev **415** will be a negative voltage. To eliminate the accumulation of the reverse recovery charge Q.sub.rr as described herein, the reverse-drain-to-reverse-source voltage should not be greater than the forward bias voltage of the body diode. The forward bias voltage of the body diode

may typically be +0.6 V. To assure that the body diode is not forward biased, the gate-to-source voltage V.sub.gs **410** minus the reverse threshold voltage V.sub.t-reverse must be large enough to switch the low switch M**2** **400** to the reverse-biased-ON state. FIG. **5** shows a representative value for the reverse source voltage S.sub.rev=−0.6 V and for the grounded reverse drain voltage D.sub.rev=0 V when the forward bias voltage of the body diode is a typical +0.6 V. Thus, the reverse-drain-to-reverse-source voltage D.sub.rev−S.sub.rev=0 V−(−0.6 V)=+0.6 V. When the overall gate-to-reverse-source voltage V.sub.bias−S.sub.rev=V.sub.bias—(—0.6 V)=V.sub.bias+0.6 V is greater than the reverse threshold voltage V.sub.t-reverse, the low switch M**2** **400** is switched to the reverse-bias-ON state and the accumulation of the reverse recovery charge Q.sub.rr is eliminated.
(40) FIG. **6** shows that the gate-to-source voltage V.sub.gs **410** creates a resistance drive R.sub.dson **445** for the low switch M**2** **400** in the reverse-bias-ON state. The product of (1) the resistance R.sub.dson **445** and (2) the current I.sub.2 **425** must be less than the forward bias voltage V.sub.f **447** of the body diode of the low switch M**2** **400** to eliminate the accumulation of the reverse recovery charge Q.sub.rr. However, even if the body diode does forward bias, only a fractional portion of the current I.sub.2 will be contributed to the reverse recovery charge Q.sub.rr, reducing the reverse recovery charge Q.sub.rr for the low switch M**2** **400** as compared to the accumulation of the reverse recovery charge Q.sub.rr for a low switch M**2** in which the turn-off voltage level is not gate controlled, such as the low switch M**2** **225** of the representative prior art power switching circuit **200**.
(41) There are several ways that this technique can be implemented into a design. FIGS. **7**A and **7**B show a Driver Stage M**3** and M**4** in series with diode D**3** which sets the V.sub.bias voltage. In addition, a parallel clamping device, M**5** can be placed to pull the gate to source voltage to zero when the drain voltage begins to rise. This also helps to mitigate the M**2** device from turning back on when the miller capacitance (Cm) from the drain to gate injects charge onto the gate during this voltage transient on the drain.
(42) FIG. **7**A shows an embodiment of the present invention, the power switching circuit **700**, including the low switch M**2** **400**. In the power switching circuit **700**, a driver stage **705** that includes the input for the driver stage input voltage V.sub.dd **710**, the driver switch M**4** **715**, the driver switch M**4** **720**, and the driver diode D**3** **725**, and the driver diode voltage V.sub.D3 **730** at the connection of the driver switch M**3** **720** and driver diode D**3** **725**. The driver stage **705** is used to set the bias voltage V.sub.bias for the low switch gate voltage V.sub.gL **407**.
(43) In the embodiment shown in FIG. **7**A, the parallel clamping device M**5** **735** is shown. The parallel clamping device M**5** **735** is used to pull the gate-to-source voltage V.sub.gL 407 to 0 V when the drain voltage of the low switch M**2** **400** begins to rise. This action helps to prevent the low switch M**2** **400** from switching to the ON state when the parasitic capacitance known as the Miller capacitance C.sub.m **740** from the drain to the gate of the low switch M**2** **400** injects charge onto the gate of the low switch M**2** **400** during the rise of the voltage at the drain. The parallel clamping device M**5** **735** may be controlled from the driver stage **705** by a delay circuit **745**.
(44) FIG. **7**B shows the behavior over time of the low switch gate voltage V.sub.gL **407** as the voltage curve **755** and of the high switch gate voltage V.sub.gH **450** as the voltage curve **760** for the embodiment shown in FIG. **7**A. The low switch gate voltage V.sub.gL **407** begins as the driver stage input voltage V.sub.dd **710**, and then the low switch gate voltage V.sub.gL **407** is changed to the driver diode voltage V.sub.D3 **730**. After a time delay Δt introduced by the delay circuit **745**, the low switch gate voltage V.sub.gL **407** is changed to 0 V. During the time delay Δt, the high switch gate voltage V.sub.gH **450** is changed to switch the high switch M**1** (not shown in FIG. **7**A) to the ON state. The parallel clamping device M**5** **735** may also be controlled by a combination of the driver stage **705** and feedback monitoring of the drain voltage of the low switch M**2** **400**.
(45) FIGS. **8**A and **8**B show a latter technique which provides immediate feedback when the drain voltage of M**2** begins to rise, which prevents inadvertent turn-off of M**2** from the C.sub.m

capacitor. FIG. **8**A shows an embodiment in which the logic gate **810** controls the gate of the parallel clamping device M**5** **735**, where the logic gate **810** uses as input (1) the voltage at the gates of the driver switches M**3** and M**4** and (2) the output of the device **812**. When the drain voltage of the low switch M**2** **400**, V.sub.sw **805**, begins to rise, the parallel clamping device M**5** is switched to the ON state, and the low switch M**2** **400** is prevented from switching to the ON state when the Miller capacitance C.sub.m **740** injects charge onto the gate of the low switch M**2** **400**. FIG. **8**A also shows the driver stage input voltage V.sub.dd **765**, the high switch **770**, the input for the high switch gate voltage V.sub.gh **775**, and the high switch drain voltage V.sub.D **780**.

(46) FIG. **8**B shows the behavior over time of the low switch gate voltage V.sub.gL **407** as the voltage curve **815**, of the drain voltage of the low switch M**2** **400**, V.sub.sw **805**, as voltage curve **820**, and of the high switch gate voltage V.sub.gH **775** as voltage curve **825**. The low switch gate voltage V.sub.L **407** begins at the driver stage input voltage V.sub.dd **765**, then is changed to the driver diode voltage V.sub.D3 **730**. The drain voltage of the low switch M**2** **400**, V.sub.sw **805**, begins at a voltage between 0 V and the representative voltage of −0.6 V and is changed to the representative voltage of −0.6 V when the low switch gate voltage V.sub.gL **407** is changed to the driver diode voltage V.sub.D3 **730**. Then the high switch gate voltage V.sub.gH **775** is changed to switch the high switch **770** to the ON state and as the high switch gate voltage V.sub.gH **775** rises, drain voltage of the low switch M**2** **400**, V.sub.sw **805**, rises. When drain voltage of the low switch M**2** **400**, V.sub.sw **805**, reaches 0 V, and the parallel clamping device M**5** is switched to the ON state. Thus, the low switch M**2** **400** is prevented from switching to the ON state when the Miller capacitance C.sub.m **740** injects charge onto the gate of the low switch M**2** **400**.

(47) Additionally, for the embodiments shown in FIGS. **7**A and **8**A, other techniques can be used where the bias voltage V.sub.bias for the low switch gate voltage V.sub.gL **407** is a dynamically adjusted value that is compensated by feedback from a current sense amplifier from the low switch M**2** **400**. This allows for greatest gate-to-reverse-source voltage drive in the reverse direction.

(48) In addition, a feedback loop can be achieved from a tap-off FET from the low switch M**2** **400** where a low-current diode voltage is monitored and the bias voltage V.sub.bias for the low switch gate voltage V.sub.gL **407** can be adjusted. In addition, improvements can be made to the technique of optimizing the threshold voltage V.sub.t of the low switch M**2** **400**. By making the threshold voltage V.sub.t of the low switch M**2** **400**, the low switch M**2** **400** can have more drive strength per unit area in the reverse-bias mode.

(49) FIG. **9**A shows an embodiment of the present invention in which a wide-bandgap low switch, such as a low switch that includes gallium nitride (Ga N), is used in a power switching circuit **900**, such as a low switch that includes gallium nitride (Ga N). FIG. **9**A shows a wide-bandgap low switch with a low switch gate voltage V.sub.gL **910**, a low switch drain voltage V.sub.dL **915**, and a low switch source voltage V.sub.sL **920**, and a high switch **925** with a high switch drain voltage V.sub.D **927** and a high switch gate voltage V.sub.gH **930**. Also shown is a load **905** including the load inductor L.sub.1 **935** and the load capacitor C.sub.1 **940**. During operation, when current flows from the low switch source **920** to the low switch drain **915** at the time during make-before-break **955**, the gate-to-source voltage V.sub.gsA is placed at a mid-level voltage Vmid **955**. By using this embodiment, the drain-to-source voltage V.sub.dsA maximum negative voltage can be minimized. Use of this embodiment aids in the reduction of noise generated.

(50) FIG. **9**B compares the behaviors over time of voltages in a power switching circuit including the wide-bandgap low switch of FIG. **9**A (part “A” of FIG. **9**B) and in a prior art power switching circuit (part “B” of FIG. **9**B). Because the maximum drain-to-source voltage of the prior switch of part “B” is greater than the maximum drain-to-source voltage of the wide-bandgap switch of FIG. **9**A, the power loss in the prior art switch of part “B” is greater than power loss of the wide-bandgap switch of FIG. **9**A, an improvement over the prior art.

(51) FIG. **10** shows that the typical silicon or silicon carbide switch is actually a four-terminal device. In the switch **1000**, having the gate G **1005**, the drain D **1010**, the source S **1015**, and the

backgate BG **1020**, the backgate **1020** is tied to the source S **1015**. The switch **1000** is equivalent to the switch **1025**, having the gate G **1030**, the drain D **1035**, the source S **1040**, and the body diode **1045**.

(52) FIG. **11**A shows an embodiment of the present invention, a four-terminal switch **1100** including silicon or silicon carbide and having the gate G **1105**, the source S **1110**, the drain D **1115**, the backgate BG **1120**, a voltage source for the gate bias voltage V.sub.bias1 **1125**, and a voltage source for the backgate bias voltage V.sub.bias2 **1130**. Both the gate bias voltage V.sub.bias1 **1125** and the backgate bias voltage V.sub.bias2 **1130** can be adjusted to further reduce the ON-state resistance of the switch **1100** when current flows from the source S **1110** to the drain D **1115** when the switch is in the reverse-bias mode. To achieve this, the gate-to-source voltage can be placed at a mid-level voltage and the backgate bias voltage V.sub.bias2 **1130** can also be placed at a mid-level voltage, which reduces the threshold voltage, increasing the drive and reducing the current of the switch **1100** with the current flowing from the source S **1110** to the drain D **1115** when the switch is in the reverse-bias mode. Any combination of the gate bias voltage V.sub.bias1 **1125** and the backgate bias voltage V.sub.bias2 **1130** can be used.

(53) FIG. **11**B shows the behavior over time of the gate bias voltage V.sub.bias1 **1125**, shown as voltage curve **1135**, and the backgate bias voltage V.sub.bias2 **1130**, shown as voltage curve **1140**, for the switch **1100** when the gate bias voltage V.sub.bias1 **1125** and the backgate bias voltage V.sub.bias2 **1130** are changed at the same times.

(54) FIG. **11**C shows the behavior over time of the gate bias voltage V.sub.bias1 **1125**, shown as voltage curve **1145**, and the backgate bias voltage V.sub.bias2 **1130**, shown as voltage curve **1150**, for the switch **1100**. This is another timing embodiment of FIG. **11**B, in which the backgate bias voltage V.sub.bias2 **1130** is activated to a voltage backgate bias voltage V.sub.bias2 **1130** prior to gate bias voltage V.sub.bias1 **1125** transitioning from a high state to a mid state. Additionally, the backgate bias voltage V.sub.bias2 **1130** transitions from a mid state to a low state after gate bias voltage V.sub.bias1 **1125** transitions from a mid state to a low state. By controlling the voltage of gate bias voltage V.sub.bias1 **1125** and backgate bias voltage V.sub.bias2 **1130** in this condition provides a stable shift in threshold voltage for the gate bias voltage V.sub.bias1 **1125** when it transitions from a high state to a mid state and from mid state to low state, including zero (0V). It is also possible to maintain backgate bias voltage V.sub.bias2 **1130** in a non-zero state, thus it does not have to dynamically switch.

(55) The approach taught herein can also be used with Silicon Carbide applications (SiC), Gallium Nitride, Gallium Oxide, or Gallium Arsenide. Any combination of bias voltage and threshold voltage can be set either positive or negative in value that meets the requirements of off-state in the forward direction and on-state in the negative direction.

(56) FIG. **12** shows an embodiment of the present invention, a power supply **1200** having a dynamic synchronous rectifier driver circuit **1202**. The power supply **1200** includes a switching transistor driver circuit **1204** connected to a switching transistor **1206**, and the synchronous rectifier driver circuit **1202** connected to a synchronous rectifier **1208**. The synchronous rectifier driver circuit **1202** and the switching transistor driver circuit **1204** are controlled by control circuit CTRL **1210**. The synchronous rectifier driver circuit **1202** will be described below in reference to FIG. **13**. The source of the switching transistor **1206** is connected to a drain of the synchronous rectifier **1208**. The drain of the switching transistor **1206** is connected to point SW. A first resister **1212** is connected to SW and in series with a diode **1214**, which is connected in series with a first capacitor **1216** connected to reference ground **1226**. The first capacitor **1216** is also connected in parallel with a switch **1218** that is series connected to a second resistor **1220**, which is connected to reference ground **1226**. The control circuit CTRL **1210** is also connected between the diode **1214** and the first capacitor **1216**. A time delay circuit **1222** is connected between the point SW and a second capacitor **1224**, which is connected to reference ground **1226**. The second capacitor **1224** is connected to the control circuit CTRL **1210**. An inductor **1228** is connected between point SW and

a voltage output Vout **1230**. A third capacitor **1232**, which is connected to the reference ground **1226**, is connected between the inductor **1228** and the voltage output Vout **1230**.
(57) FIG. **13** shows an embodiment of the present invention, the dynamic synchronous rectifier driver circuit **1202** of FIG. **12**. A first transistor **1302** has a gate connected to the control circuit CTRL **1210** and a drain connected to the gate LSG of the synchronous rectifier MD **1208**. A second transistor **1304** has a drain connected to the gate LSG of the synchronous rectifier MD **1208**, a gate connected to the control circuit CTRL **1210**, and a source connected to the output Vm of a first amplifier **1306**. A third transistor **1308** has a drain connected to the gate LSG of the synchronous rectifier **1208**, a gate connected to the control circuit CTRL **1210** and a source connected to reference ground **1322**. A voltage divider **1310** includes a current source ILK-REF **1312** and a fourth transistor MD/x **1314** with an output into an input of a second amplifier **1316**. The fourth transistor MD/x **1314** has a gate connected to its drain and a source connected to the reference ground **1322**. The fourth transistor MD/x the same type, but proportionally smaller than, the synchronous rectifier MD **1208**. The output of the second amplifier **1316** is connected an input of the first amplifier **1306** and a first resistor **1318** that is series connected to a variable resistor **1320** and the reference ground **1322**. The control circuit CTRL **1210** controls the variable resistor **1320** to set the synchronous rectifier voltage driving level.
(58) FIG. **14** shows an embodiment of the present invention, signal timing diagrams for the power supply of FIG. **12**. The switching transistor gate drive voltage V.sub.HSG is high and the synchronous rectifier gate voltage V.sub.LSG is low at time **1402**. At time **1404**, switching transistor gate drive voltage V.sub.HSG goes low. After a dead time DT, the synchronous rectifier gate voltage V.sub.LSG goes high. At time **1408**, the synchronous rectifier gate voltage V.sub.LSG drops to a dynamically adjustable, intermediate turn off voltage V.sub.m and remains at that level for the drop time DT. At time **1410**, the synchronous rectifier gate voltage V.sub.LSG remains at V.sub.m and the switching transistor gate drive voltage V.sub.HSG goes high. After a dynamically adjustable hold time t.sub.m, the synchronous rectifier gate voltage V.sub.LSG goes low at time **1412**.
(59) The following embodiments reduce or eliminate current or voltage transients in the switching circuit by actively monitoring the SW pin located between the switching transistor and the synchronous rectifier. The circuits use closed loop feedback to control level transistion(s) for the synchronous rectifier gate. The active feedback may include a voltage, a rate of change of voltage, a current, a rate of change of current or a combination thereof, all at the SW pin. The circuit does not require, but may use, signals from or monitoring of the switching transistor. Moreover, the circuit does not require, but may include, a time delay circuit. As a result, the circuits described herein reduce current losses from break before make, gate to drain capacitance, reverse recovery charge (Q.sub.rr) and gate bounce. The circuits also help prevent ringing.
(60) FIG. **15** shows an embodiment of the present invention, a flow chart of a method **1500** for controlling a synchronous rectifier. The synchronous rectifier is turned OFF in block **1502** by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero. One or more values of a drain or a source of the synchronous rectifier are monitored in block **1504**. The gate of the synchronous rectifier is driving to a second voltage that is less than the one or more first voltages in block **1506** whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below one or more threshold values.
(61) In one aspect, a switching circuit includes the synchronous rectifier connected to a switching transistor, and the method further includes reducing or eliminating current or voltage transients in the switching circuit. In another aspect, the method further includes selecting the one or more first voltages or the second voltage. In another aspect, the one or more first voltages are less than a threshold voltage of the synchronous rectifier. In some embodiments, the one or more first voltages can be greater than or equal to the threshold voltage of the synchronous rectifier. In another aspect, the one or more first voltages comprise a set of descending stair-step values, or a set of descending

values forming a ramp or a curve. In another aspect, the one or more values can be a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier. In another aspect, the method further includes adjusting the one or more threshold values or a delay in driving the gate of the synchronous rectifier to the second voltage using a predictive feedback process. In another aspect, the predictive feedback process: samples and stores the one or more values of the gate, drain or the source of the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the one or more threshold values. In another aspect, the predictive feedback process: samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the one or more threshold values. In another aspect, the predictive feedback process: samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. In another aspect, the predictive feedback process: samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage.

(62) In another aspect, the second voltage is equal to zero volts. In another aspect, the second voltage is less than zero volts, e.g., −1 volts, −2 volts or an incremental value between 0 and −2 volts or more. In another aspect, a gate of a switching transistor connected to the synchronous rectifier is not monitored or used to control the gate of the synchronous rectifier. In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a driver circuit connected to the gate of the synchronous rectifier; a parallel clamping circuit connected to the gate of the synchronous rectifier; a monitoring circuit connected to the drain or the source of the synchronous rectifier and the parallel clamping circuit; and a control circuit connected to the driver circuit and the monitoring circuit. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the driver circuit include: a first transistor having a gate connected to the control circuit and a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a gate connected to the control circuit; a one-way current switch connected to a source of the second transistor. In another aspect, the parallel clamping circuit includes a third transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the monitoring circuit and a source connected to a reference ground. In another aspect, the monitoring circuit include: a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier, a gate connected to the control circuit, and a source connected to the parallel clamping circuit; and a fifth transistor having a drain connected to the parallel clamping circuit and a gate connected to the control circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the driver circuit, the parallel clamping circuit, the monitoring circuit and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

(63) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connected between a source of the second transistor and a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor

having a drain connected to the gate of the third transistor and a source connected to the reference ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the one or more first voltages by: turning the first transistor from ON to OFF; turning the second transistor from OFF to ON; turning the fourth transistor from OFF to ON; and turning the fifth transistor from ON to OFF. In another aspect, the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the second voltage by turning the third transistor from OFF to ON. In another aspect, the synchronous rectifier driver circuit includes or does not include a time delay circuit. In another aspect, the control circuit is further connected to a modulation signal. In another aspect, the first transistor, the second transistor, the one-way current switch, the third transistor, the fourth transistor, the fifth transistor and the control circuit are formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

(64) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a one-way current switch connected from a source to the drain of the sixth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.

(65) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier; a second one-way current switch connected from a source to the drain of the fourth transistor; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(66) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connected to a source of the second transistor; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current switch and a source connected to a reference ground; a second one-way current switch connected between a voltage and the gate of the fourth transistor; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the

reference ground; a seventh transistor having a drain connected to the drain or the source of the synchronous rectifier; a third one-way current switch connected from a source to the drain of the seventh transistor; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

(67) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier; a second one-way current switch connected from a source to the drain of the fourth transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(68) In another aspect, the synchronous rectifier is controlled using a synchronous rectifier driver circuit including: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

(69) FIG. **16** shows an embodiment of the present invention, a power converter **1600**. The power converter **1600** includes a switching transistor **1602**, a switching transistor driver circuit **1604** connected to a gate of the switching transistor **1602**, a synchronous rectifier **1606** connected to the switching transistor **1602**. Pin SW is connected to the source or drain of the synchronous rectifier **1606**. A synchronous rectifier driver circuit **1608** is connected to a gate of the synchronous rectifier **1606** and the source or drain of the synchronous rectifier **1606**. The synchronous rectifier driver circuit **1608** is configured to: (1) turn the synchronous rectifier **1606** OFF by driving the gate of the synchronous rectifier **1606** to one or more first voltages that are non-zero; (2) monitor one or more values of the drain or the source of the synchronous rectifier **1606**; and (3) drive the gate of the synchronous rectifier **1606** to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier **1606** pass above or below one or more threshold values. The synchronous rectifier driver circuit **1608** reduces or eliminates current or voltage transients in the power converter **1600**. In some embodiment, the one or more first voltages can be less than a threshold voltage of the synchronous rectifier **1606**. In other embodiments, the one or more first voltages can be greater than or equal to the threshold voltage of the synchronous rectifier. The one or more first voltages comprise a set of descending stair-step values, or a set of descending values forming a ramp or a curve. The second voltage can be equal to zero volts, or less than zero volts, e.g., −1 volts, −2 volts or an incremental value between 0 and −2 volts or more. In some embodiments, the gate of the switching transistor **1602** is not monitored or used to control the gate of the synchronous rectifier. In one aspect, the one or

more values can be a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier.

(70) In other embodiments, the one or more threshold values or a delay in driving the gate of the synchronous rectifier **1606** to the second voltage are adjusted using a predictive feedback process. Since most switching regulators and intelligent motion drivers are periodic, the loop can be adjusted by increasing or decreasing phase delay. The current, voltage, rate of change of the current or voltage, or a combination thereof can be sampled and stored to be used for the next cycle. This can also be used as a running average over several previous cycles. This predictive feedback process can sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the one or more threshold values.

(71) In addition to sensing the drain or source synchronous rectifier **1606**, the switching transistor can be sensed as well. This predictive feedback process can sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the one or more threshold values.

(72) In another embodiment, the closed loop system can be put on a delay to turn off the synchronous rectifier **1606**. This can be increased or decreased in the same manner by sampling the current, voltage or rate of change and adjusting time delay appropriately. This predictive feedback process can sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. Similarly, this predictive feedback process can sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier, and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage. Note that combinations of the previously describe predictive feedback processes can be used

(73) As shown in FIG. **16**, the synchronous rectifier driver circuit **1608** includes a driver circuit **1610** connected to the gate of the synchronous rectifier **1606**, a parallel clamping circuit **1612** connected to the gate of the synchronous rectifier **1606**, a monitoring or sensing circuit connected to the drain or the source of the synchronous rectifier **1606** and the parallel clamping circuit **1612**, and a control circuit **1616** connected to the driver circuit **1610** and the monitoring or sensing circuit **1614**. The synchronous rectifier driver circuit **1608** may or may not include a time delay circuit as previously described.

(74) In some embodiments, such as FIG. **17**, the driver circuit **1610** includes a first transistor **1702** having a gate connected to the control circuit **1616** and a drain connected to the gate of the synchronous rectifier **1606**, a second transistor **1704** having a drain connected to the gate of the synchronous rectifier **1606** and a gate connected to the control circuit **1616**, and a one-way current switch **1706** connected to a source of the second transistor. The one-way current switch **1706** can be a diode, a MOS connected diode, an attenuated MOS connected diode, a current controlled scaled down power FET with low impedance buffer, made from a power FET, or a low impedance buffered voltage reference, etc. The parallel clamping circuit **1612** includes a third transistor **1708** having a drain connected to the gate of the synchronous rectifier **1606**, a gate connected to the monitoring circuit **1614** and a source connected to a reference ground **1618**. The monitoring or sensing circuit **1614** includes a fourth transistor M**1 1710** having a drain connected to the drain or the source of the synchronous rectifier **1606**, a gate connected to the control circuit **1616** and a source connected to the parallel clamping circuit **1612**, and a fifth transistor M**2 1712** having a drain connected to the parallel clamping circuit **1612** and a gate connected to the control circuit

**1616**. The control circuit **1616** is further connected to a modulation signal PWM. The driver circuit **1610**, the parallel clamping circuit **1612**, the monitoring circuit **1614** and the control circuit **1616** can be formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.
(75) FIG. **17** shows an embodiment of the present invention, a schematic diagram of a circuit **1700** for controlling a synchronous rectifier MD **1606** having a source or a drain connected to a switching transistor (see FIG. **16**) and pin SW. The synchronous rectifier driver circuit **1608** includes: a first transistor **1705** having a drain connected to the gate LSG of the synchronous rectifier MD **1606**; a second transistor **1704** having a drain connected to the gate LSG of the synchronous rectifier MD **1606**; a one-way current switch **1706** connected between a source of the second transistor **1704** and a reference ground **1714**; a third transistor **1708** having a drain connected to the gate LSG of the synchronous rectifier **1606** and a source connected to the reference ground **1714**; a fourth transistor M**1 1710** having a drain connected to the drain or the source of the synchronous rectifier MD **1606** and a source connected to a gate of the third transistor **1708**; a fifth transistor M**2 1712** having a drain connected to the gate of the third transistor **1708** and a source connected to the reference ground **1714**; and a control circuit **1616** connected to a gate of the first transistor **1702**, a gate of the second transistor **1704**, a gate of the fourth transistor M**1 1710** and a gate of the fifth transistor M**2 1712**.
(76) The synchronous rectifier driver circuit **1608** drives the gate of the synchronous rectifier MD **1606** to the one or more first voltages by: turning the first transistor **1702** from ON to OFF; turning the second transistor **1704** from OFF to ON; turning the fourth transistor M**1 1710** from OFF to ON; and turning the fifth transistor M**2 1712** from ON to OFF. The synchronous rectifier driver circuit **1608** drives the gate of the synchronous rectifier MS **1606** to the second voltage by turning the third transistor **1708** from OFF to ON when a positive voltage is observed on the SW pin. The synchronous rectifier driver circuit **1608** may or may not include a time delay circuit. In addition, the control circuit **1616** is further connected to a modulation signal. The first transistor, the second transistor, the one-way current switch, the third transistor, the fourth transistor, the fifth transistor and the control circuit can be formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.
(77) FIG. **18** shows an embodiment of the present invention, a schematic diagram of a circuit **1800** for controlling a synchronous rectifier MD **1606** connected to switching transistor (see FIG. **16**) and pin SW. The synchronous rectifier driver circuit **1608** includes: a first transistor **1705** having a drain connected to the gate LSG of the synchronous rectifier MD **1606**; a second transistor **1704** having a drain connected to the gate LSG of the synchronous rectifier MD **1606**; a circuit **1802** for dynamically adjusting a turn off voltage for the synchronous rectifier MD **1606** (see e.g., FIGS. **12-14**) is connected between a source of the second transistor **1704** and a reference ground **1714**; a third transistor **1708** having a drain connected to the gate LSG of the synchronous rectifier **1606** and a source connected to the reference ground **1714**; a fourth transistor M**1 1710** having a drain connected to the drain or the source of the synchronous rectifier MD **1606** and a source connected to a gate of the third transistor **1708**; a fifth transistor M**2 1712** having a drain connected to the gate of the third transistor **1708** and a source connected to the reference ground **1714**; and a control circuit **1616** connected to a gate of the first transistor **1702**, a gate of the second transistor **1704**, a gate of the fourth transistor M**1 1710** and a gate of the fifth transistor M**2 1712**.
(78) FIG. **19** shows an embodiment of the present invention, a schematic diagram **1900** of a synchronous rectifier driver circuit **1902** connected to a synchronous rectifier **1606** having a body diode **1904**. The synchronous rectifier driver circuit **1902** includes: a first transistor **1702** having a drain connected to the gate of the synchronous rectifier **1606**; a second transistor **1906** having a drain connected to the gate of the synchronous rectifier **1606** and a body connected to a reference ground **1922**; a third transistor **1908** having a drain connected to a source of the second transistor **1906**, and a body and a drain connected to the reference ground **1922**; a fourth transistor **1910** having a drain connected to the gate of the synchronous rectifier **1606**, a gate connected to a source

of the second transistor **1906** and a source connected to the reference ground **1922**; a fifth transistor **1912** having a drain connected to the gate of the synchronous rectifier **1606** and a source connected to the reference ground **1922**; a sixth transistor **1914** having a body diode **1916**, a drain connected to the drain or the source of the synchronous rectifier **1606** and a source connected to the gate of the fifth transistor **1912**; a seventh transistor **1918** having a drain connected to the gate of the fifth transistor **1912** and a source connected to the reference ground **1922**; an inverter **1920** having an output connected to the gates of the first transistor **1702**, the second transistor **1906** and the sixth transistor **1914**; and a control signal CTRL connected to the input of the inverter **1920** and the gates of the third transistor **1908** and seventh transistor **1918**.

(79) FIG. **20** shows an embodiment of the present invention, a schematic diagram **2000** of a synchronous rectifier driver circuit **2002** connected to a synchronous rectifier **1606** having a body diode **1904**. The synchronous rectifier driver circuit **2002** includes: a first transistor **1702** having a drain connected to the gate of the synchronous rectifier **1606**; a second transistor **1704** having a drain connected to the gate of the synchronous rectifier **1606**; a one-way current switch **1706** connecting a source of the second transistor **1704** to a reference ground **2004**; a third transistor **1912** having a drain connected to the gate of the synchronous rectifier **1606** and a source connected to the reference ground **2004**; a fourth transistor **1914** having a body diode **1916** and a drain connected to the drain or the source of the synchronous rectifier **1606**; an amplifier **2006** having a positive terminal (+) connected to the source of the fourth transistor **1914**, a negative terminal (−) connected to a voltage V.sub.x (−0.3 V<V.sub.x<0 V); and an output connected to the gate of the third transistor **1912**; a fifth transistor **1918** having a drain connected to the gate of the third transistor **1912** and a source connected to the reference ground **2004**; an inverter **1920** having an output connected to the gates of the first transistor **1702**, the second transistor **1704**, an enable terminal of the amplifier **2006** and the fourth transistor **1914**; and a control signal CTRL connected to an input of the inverter **1920** and the gate of the fifth transistor **1918**.

(80) FIG. **21** shows an embodiment of the present invention, a schematic diagram **2100** of a synchronous rectifier driver circuit **2102** connected to a synchronous rectifier **1606** having a body diode **1904**. The synchronous rectifier driver circuit **2102** includes: a first transistor **1702** having a drain connected to the gate of the synchronous rectifier **1606**; a second transistor **1704** having a drain connected to the gate of the synchronous rectifier **1606**; a first one-way current switch **1706** connected to a source of the second transistor **1704**; a fourth transistor **1910** having a drain connected to the gate of the synchronous rectifier **1606**, a gate connected to the first one-way current switch **1706** and a source connected to a reference ground **2104**; a second one-way current switch **2106** connected between a voltage V.sub.x and the gate of the fourth transistor **1910**; a fifth transistor **2108** having a drain connected to the gate of the fourth transistor **1910** and a source connected to the reference ground **2104**; a sixth transistor **1912** having a drain connected to the gate of the synchronous rectifier **1606** and a source connected to the reference ground **2104**; a seventh transistor **1914** having a body diode **1916** and a drain connected to the drain or the source of the synchronous rectifier **1606**; an inverter **1920** having an output connected to the gates of the first transistor **1702**, the second transistor **1704** and the seventh transistor **1914**; and a control signal CTRL connected to an input of the inverter **1920** and the gate of the fifth transistor **2108**, the gate of the sixth transistor **1912** and the source of the seventh transistor **1914**.

(81) FIG. **22** shows an embodiment of the present invention, a schematic diagram **2200** of a synchronous rectifier driver circuit **2202** connected to a synchronous rectifier **1606** having a body diode **1904**. The synchronous rectifier driver circuit **2202** includes: a first transistor **1702** having a drain connected to the gate of the synchronous rectifier **1606**; a second transistor **1704** having a drain connected to the gate of the synchronous rectifier **1606**; a one-way current switch **1706** connecting a source of the second transistor **1704** to a reference ground **2204**; a third transistor **1912** having a drain connected to the gate of the synchronous rectifier **1606** and a source connected to the reference ground **2204**; a fourth transistor **1914** having a body diode **1916** and a drain

connected to the drain or the source of the synchronous rectifier **1606**; a fifth transistor **1918** having a drain connected to the gate of the third transistor **1912** and a source connected to the reference ground **2204**; a sixth transistor **2206** having a drain connected to the source of the first transistor **1702**, a gate connected to the source of the fourth transistor **1914** and a source connected to a gate of the third transistor **1912**; an inverter **1920** having an output connected to the gates of the first transistor **1702**, the second transistor **1704**, and the fourth transistor **1914**; and a control signal CTRL connected to an input of the inverter **1920** and the gate of the fifth transistor **1918**.

(82) FIG. **23** shows an embodiment of the present invention, a schematic diagram **2300** of a synchronous rectifier driver circuit **2002** connected to a synchronous rectifier **1606** having a body diode **1904**. The synchronous rectifier driver circuit **2002** includes: a first transistor **1702** having a drain connected to the gate of the synchronous rectifier **1606**; a second transistor **1704** having a drain connected to the gate of the synchronous rectifier **1606**; a first one-way current switch **1706** connecting a source of the second transistor **1704** to a reference ground **2004**; a third transistor **1912** having a drain connected to the gate of the synchronous rectifier **1606** and a source connected to the reference ground **2004**; a fourth transistor **1914** having a body diode **1916**, a drain connected to the drain or the source of the synchronous rectifier **1606**, a source connected to the gate of the third transistor **1912**; a fifth transistor **1918** having a drain connected to the gate of the third transistor **1912** and a source connected to the reference ground **2004**; an inverter **1920** having an output connected to the gates of the first transistor **1702**, the second transistor **1704**, an enable terminal of the amplifier **2006** and the fourth transistor **1914**; and a control signal CTRL connected to an input of the inverter **1920** and the gate of the fifth transistor **1918**.

(83) The synchronous rectifier driver circuits shown FIGS. **16-23** can be operated using the method described in reference to FIG. **15**. Note also that the operation of the synchronous rectifier driver circuits shown in FIGS. **18-23** is similar to that described in reference to FIG. **17**. In addition, the synchronous rectifier driver circuits shown FIGS. **16-23** can be use to drive synchronous rectifiers within any power supply or power converter.

(84) FIGS. **24-26** depict various non-limiting signal diagrams of a buck switching regulator. In each simulation, a Vin of 12V, Vout of 1V, Iload of **20**A and a switching frequency of 1 Mhz was run. The mode of operation that is depicted in the waveforms is where the output requires charging therefore, the low side needs to turn off and the high-side turn on. The three phases of this are: (1) low-side ON/high-side OFF then low-side goes from ON to OFF; (2) low-side OFF; and finally (3) high-side ON. To maintain high efficiency, phase 2 is a very short period, typically ranging from 2 ns to 10 ns.

(85) FIG. **24** depicts signal diagrams of a buck switching regulator in accordance with the prior art. FIG. **24** shows that during phase 2 the Vgate (LS) goes immediately from ON to strong OFF. Once this happens it will cause a build-up of charge in the low-side body diode, as depicted in Q.sub.rr(LS). Due to inherent parasitics in the power lines the switch voltage translates this larger current spike into a large voltage spike on the switch pin, as depicted on V.sub.sw waveform, which is noted as Vdelta and in this case is 15V. In this example, the breakdown voltage of the low-side has been met. In order to avoid this, it would require a low-side switch to have a breakdown voltage in excess of 27V. A larger breakdown voltage low side will translate to inefficiencies in conduction loss or die area penalty.

(86) FIG. **25** depicts signal diagrams of a buck switching regulator in accordance with the prior art. FIG. **25** shows that during phase 2 for Vgate (LS), vmid voltage and Strong OFF (0V) are used. However, this case shows that there is error time (delta t-error). This error is time delay error associated from going from Vmid to Strong OFF vs turning the HS from OFF to ON. Any error time will cause a build-up of charge in the low-side body diode, as depicted in Q.sub.rr (LS). This Q.sub.rr buildup translates into a larger current spike seen on Idrain (LS) and Isource (LS). Due to inherent parasitics in the power lines the switch voltage translates this larger current spike into a large voltage spike on the switch pin, as depicted on V.sub.sw waveform, which is noted as Vdelta

and in this case is 15V. In this example, the breakdown voltage of the low-side has been met. In order to avoid this, it would require a low-side switch to have a breakdown voltage in excess of 27V. A larger breakdown voltage low side will translate to inefficiencies in conduction loss or die area penalty.
(87) FIG. **26** depicts signal diagrams of a buck switching regulator in accordance with the present invention using an active feedback circuit. FIG. **26** shows that during phase 2 for Vgate (LS), Vmid voltage and Strong OFF (0V) are used. However, a feedback mechanism is used (Vfeedback). It is observing the voltage of the switch pin. This is shown in the waveform where both the switch pin and the feedback pin are super-imposed. (V.sub.sw/Vfeeback). This feedback causes the gate voltage is transitioned to Strong OFF, which provides accurate timing turn-off of the low-side when the high-side is turning on. As result of this it can be observed that no appreciable charge build is observed on the low-side body diode, Q.sub.rr (LS). In addition, the current transients of the drain and source currents are significantly reduced. This finally translates into the voltage transients observed on the switch pin to show a delta voltage increase of only 2V. In the example the low side device requires a much lower breakdown voltage and allows for use of a more efficient device which can have much lower conduction losses as well as overall higher power efficiency when compared to prior art schemes.
(88) The approach taught herein can also be used in any combination of configurations, such as high-side drive, low-side drive, high- and low-side, or one or more FETs in any combination.
(89) The approach taught herein can also be used in any combination of current or voltage switching or current and voltage switching, such as power regulation, one or more switches for line termination, DC motor drivers, induction motor drivers, transducer drivers, solid-state fuse switches, battery management, AC-to-DC power conversion, DC-to-AC power conversion, or power correction.
(90) Any combination of bias voltage and threshold voltage can be set either positive or negative in value that meets the requirements of off-state in the forward direction and on-state in the negative direction.
(91) Any combination of transitions from on voltage to bias voltage that meets the requirements of on- to off-state in the forward direction and on-state in the negative direction is included in the present invention. The transitions can also be either step functions, linear graded, or non-linear graded transitions.
(92) It will be understood that particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.
(93) All publications and patent applications mentioned in the specification are indicative of the level of skill of those skilled in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.
(94) The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the device, the method being employed to determine the value, or the variation that exists among the study subjects.

(95) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps. In embodiments of any of the compositions and methods provided herein, “comprising” may be replaced with “consisting essentially of” or “consisting of.” As used herein, the phrase “consisting essentially of” requires the specified integer(s) or steps as well as those that do not materially affect the character or function of the claimed invention. As used herein, the term “consisting” is used to indicate the presence of the recited integer (e.g., a feature, an element, a characteristic, a property, a method/process step, or a limitation) or group of integers (e.g., feature(s), element(s), characteristic(s), property(ies), method/process(s) steps, or limitation(s)) only.
(96) The term “or combinations thereof” as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof” is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, AB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.
(97) As used herein, words of approximation such as, without limitation, “about,” “substantial” or “substantially” refers to a condition that when so modified is understood to not necessarily be absolute or perfect but would be considered close enough to those of ordinary skill in the art to warrant designating the condition as being present. The extent to which the description may vary will depend on how great a change can be instituted and still have one of ordinary skill in the art recognize the modified feature as still having the required characteristics and capabilities of the unmodified feature. In general, but subject to the preceding discussion, a numerical value herein that is modified by a word of approximation such as “about” may vary from the stated value by at least ±1, 2, 3, 4, 5, 6, 7, 10, 12 or 15%.
(98) As used herein, the term “connected” refers to devices or components that are directly or indirectly connected or coupled together. One or more other devices or components can be disposed between the connected devices or components.
(99) All of the devices and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the devices and/or methods of this invention have been described in terms of particular embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.
(100) Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope and spirit of the disclosure. Accordingly, the protection sought herein is as set forth in the claims below.
(101) Modifications, additions, or omissions may be made to the systems and apparatuses described herein without departing from the scope of the invention. The components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses may be performed by more, fewer, or other components. The methods may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order.

(102) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants wish to note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words "means for" or "step for" are explicitly used in the particular claim.

## Claims

1. A method for controlling a synchronous rectifier comprising: turning the synchronous rectifier OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero; monitoring one or more values of a drain or a source of the synchronous rectifier; driving the gate of the synchronous rectifier to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below a threshold value comprising zero; and adjusting the threshold value or a delay in driving the gate of the synchronous rectifier to the second voltage using a predictive feedback process.
2. The method of claim 1, wherein: a switching circuit comprises the synchronous rectifier connected to a switching transistor; and reducing or eliminating current or voltage transients in the switching circuit.
3. The method of claim 1, wherein: the one or more first voltages are less than a threshold voltage of the synchronous rectifier; or the one or more first voltages comprise a set of descending stair-step values, or a set of descending values forming a ramp or a curve; or the one or more values comprise a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier.
4. The method of claim 1, wherein the predictive feedback process: samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the threshold value.
5. The method of claim 1, wherein the predictive feedback process: samples and stores the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier; and uses the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the threshold value.
6. The method of claim 1, wherein the second voltage is less than or equal to zero volts.
7. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connected between a source of the second transistor and a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor.
8. The method of claim 7, wherein the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the one or more first voltages by: turning the first transistor from ON to OFF; turning the second transistor from OFF to ON; turning the fourth transistor from OFF to ON; and turning the fifth transistor from ON to OFF.
9. The method of claim 7, wherein the synchronous rectifier driver circuit drives the gate of the synchronous rectifier to the second voltage by turning the third transistor from OFF to ON.
10. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous

rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a body diode, a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.

11. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

12. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connected to a source of the second transistor; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current switch and a source connected to a reference ground; a second one-way current switch connected between a voltage and the gate of the fourth transistor; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a seventh transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

13. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a body diode and a drain connected to the drain or the source of the synchronous rectifier; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth

transistor.
14. The method of claim 1, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.
15. A circuit comprising: a synchronous rectifier driver circuit configured to: turn a synchronous rectifier OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero, monitor one or more values of a drain or a source of the synchronous rectifier, and drive the gate of the synchronous rectifier to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below a threshold value comprising zero; and adjust the threshold value or a delay in driving the gate of the synchronous rectifier to the second voltage using a predictive feedback process.
16. The circuit of claim 15, wherein: the one or more first voltages are less than a threshold voltage of the synchronous rectifier; or the one or more first voltages comprise a set of descending stair-step values, or a set of descending values forming a ramp or a curve; or the one or more values comprise a voltage, a change in the voltage, a current, a change in the current, or a combination of the voltage, the change in the voltage, the current or the change in the current of the drain or the source of the synchronous rectifier.
17. The circuit of claim 15, wherein the predictive feedback process is configured to: sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier; and use the stored values of the gate, the drain or the source of the synchronous rectifier to adjust the delay in driving the gate of the synchronous rectifier to the second voltage.
18. The circuit of claim 15, wherein the predictive feedback process is configured to: sample and store the one or more values of the gate, the drain or the source of the synchronous rectifier and a switching transistor connected to the synchronous rectifier; and use the stored values of the gate, the drain or the source of the synchronous rectifier and the switching transistor to adjust the delay in driving the gate of the synchronous rectifier to the second voltage.
19. The circuit of claim 15, wherein the second voltage is equal to or less than zero volts.
20. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a driver circuit connected to the gate of the synchronous rectifier; a parallel clamping circuit connected to the gate of the synchronous rectifier; a monitoring circuit connected to the drain or the source of the synchronous rectifier and the parallel clamping circuit; and a control circuit connected to the driver circuit and the monitoring circuit.
21. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connected between a source of the second transistor and a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to a gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference

ground; and a control circuit connected to a gate of the first transistor, a gate of the second transistor, a gate of the fourth transistor and a gate of the fifth transistor.

22. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier and a body connected to a reference ground; a third transistor having a drain connected to a source of the second transistor, and a body and a drain connected to the reference ground; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to a source of the second transistor and a source connected to the reference ground; a fifth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a sixth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the fifth transistor; a one-way current switch connected from a source to the drain of the sixth transistor; a seventh transistor having a drain connected to the gate of the fifth transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor and the sixth transistor; and a control signal connected to an input of the inverter and the gates of the third transistor and seventh transistor.

23. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier; a second one-way current switch connected from a source to the drain of the fourth transistor; an amplifier having a positive terminal connected to the source of the fourth transistor, a negative terminal connected to a voltage, and an output connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

24. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connected to a source of the second transistor; a fourth transistor having a drain connected to the gate of the synchronous rectifier, a gate connected to the first one-way current switch and a source connected to a reference ground; a second one-way current switch connected between a voltage and the gate of the fourth transistor; a fifth transistor having a drain connected to the gate of the fourth transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a seventh transistor having a drain connected to the drain or the source of the synchronous rectifier; a third one-way current switch connected from a source to the drain of the seventh transistor; an inverter having an output connected to the gates of the first transistor, the second transistor and the seventh transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor, the gate of the sixth transistor and the source of the seventh transistor.

25. The circuit of claim 15, wherein the synchronous rectifier driver circuit comprises: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a first one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier; a

second one-way current switch connected from a source to the drain of the fourth transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; a sixth transistor having a drain connected to the source of the first transistor, a gate connected to the source of the fourth transistor and a source connected to a gate of the third transistor; an inverter having an output connected to the gates of the first transistor, the second transistor, and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

26. The circuit of claim 15, wherein the synchronous rectifier is controlled using a synchronous rectifier driver circuit comprising: a first transistor having a drain connected to the gate of the synchronous rectifier; a second transistor having a drain connected to the gate of the synchronous rectifier; a one-way current switch connecting a source of the second transistor to a reference ground; a third transistor having a body diode, a drain connected to the gate of the synchronous rectifier and a source connected to the reference ground; a fourth transistor having a drain connected to the drain or the source of the synchronous rectifier and a source connected to the gate of the third transistor; a fifth transistor having a drain connected to the gate of the third transistor and a source connected to the reference ground; an inverter having an output connected to the gates of the first transistor, the second transistor, an enable terminal of the amplifier and the fourth transistor; and a control signal connected to an input of the inverter and the gate of the fifth transistor.

27. The circuit of claim 15, wherein the synchronous rectifier driver circuit is formed in or on silicon, silicon carbide, gallium nitride, gallium oxide, or gallium arsenide.

28. A power converter comprising: a switching transistor; a switching transistor driver circuit connected to a gate of the switching transistor; a synchronous rectifier connected the switching transistor; and a synchronous rectifier driver circuit connected to a gate of the synchronous rectifier and a drain or a source of the synchronous rectifier, wherein the synchronous rectifier driver circuit is configured to: turn a synchronous rectifier OFF by driving a gate of the synchronous rectifier to one or more first voltages that are non-zero, monitor one or more values of a drain or a source of the synchronous rectifier, and drive the gate of the synchronous rectifier to a second voltage that is less than the one or more first voltages whenever the one or more values of the drain or the source of the synchronous rectifier pass above or below a threshold value comprising zero, and adjust the threshold value or a delay in driving the gate of the synchronous rectifier to the second voltage using a predictive feedback process.